Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI/ML Engineer · Data Engineer · Agentic AI · LLM Systems · Python · LangGraph · RAG · LLM Evaluation · Healthcare AI · AWS · Azure · GCP · CI/CD · Production AI

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Virginia, USA — Open to Remote & On-site

PROFESSIONAL SUMMARY

Senior Software Engineer and AI/ML Engineer with 8+ years owning production systems in healthcare software and 2+ years hands-on agentic AI and LLM delivery. At Entrata: **long-term owner of HUD compliance pipelines** (few thousand federal rows → 180,000 domain records) powering eligibility, rent, and tax-credit calculations for 100K+ housing units at 99.5% uptime — the kind of work where a wrong import affects thousands of residents' lives. Built **Leasel**: 8-agent LangGraph compliance investigation platform with HITL, confidence scoring, and deterministic audit trail (93.6% precision, 87.9% F1 across 1,240 lease PDFs). Built **AgentProbe**: open-source AI evaluation toolkit with 5-metric eval engine, pass@k harness, CI/CD regression gate, and model routing. Built **AI on FHIR**: NLP clinical platform over FHIR EHR APIs with HIPAA-compliant RBAC. Strong across full AI/ML stack: agentic systems, RAG pipelines, LLM evaluation, distributed backend systems, cloud infrastructure (AWS, Azure, GCP), and data engineering. Currently building ML models and pipelines at Blue Horse Digital. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Developing ML models and data analysis pipelines for software applications and business intelligence projects — data collection, cleaning, feature engineering, model training, evaluation, and integration with back-end and front-end teams via application-facing APIs and reporting layers.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · LangGraph · RAG · Compliance Systems

- Built agentic mission-planning system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) with RAG over FAA Part 107 regulatory corpus, function calling, structured outputs, and validation guardrails enforcing compliance rules. Full ownership: ideation to deployed production client system.
- Parsed natural-language survey requests into structured flight plans using quantitative photogrammetry math; citation-grounded every compliance check to source regulation; blocked non-compliant missions deterministically before MAVLink/QGroundControl export.

George Mason University, College of Engineering and Computing
Aug 2024 – Aug 2025

Research Associate & Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) on WHO + UK Road Safety datasets across 190+ countries — $R^2 = 0.977$; applied Isolation Forest anomaly detection across 3 research pipelines; fine-tuned custom classification model (HuggingFace Transformers + PyTorch), deployed inference API on Quart (async Python); designed Tableau, ggplot2, and Quarto dashboards.
- Teaching Assistant for 600+ students: Python programming (OOP, data structures, automation scripting), database design (ER modeling, SQL, MS Access); one-on-one debugging support and rubric-aligned grading.

Entrata (PropTech / Real Estate SaaS — 100K+ MAU, 99.5% Uptime)
Sep 2018 – Aug 2024

Senior Software Engineer · Lehi, UT (Remote) — Python · PHP · PostgreSQL · RabbitMQ · AWS · Docker

- **Owned HUD income-limit import pipeline for 7 years** — few thousand federal source rows expanded into 180,000 domain records powering eligibility, rent (HAP), and tax-credit calculations for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation. Zero compliance violations across the entire ownership period.
- **Built and owned RabbitMQ-based async processing system** for gross-rent updates across thousands of tenant records — retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual rent-schedule cycles.
- **Optimized 200+ SQL queries (30% performance gain)** via execution plan analysis, indexing, partitioning, caching, and stored procedure refactoring across complex multi-tenant schemas for eligibility, financial, and document workflows.
- **Led AWS S3 document migration with 100% PHI compliance** — Libsodium-encrypted pipelines, zero security incidents; led microservices redesign using Docker — 50% scalability improvement, 40% deployment time reduction. Shipped 20+ full-stack production modules across 120+ Agile sprints.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools that reduced debugging time 40% and kept bug count below 10 for 6+ consecutive months.

Entrata India, Pune
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — PHP · Python · Java · PostgreSQL · AWS S3

- Built property management and affordable housing modules; led AWS S3 cloud migration resolving 90% of legacy storage constraints; redesigned document storage database architecture migrating hundreds of thousands of records; created 56 state-specific housing certification documents.

PROJECTS

Leasel: 8-Agent LangGraph Compliance Platform — HITL, Deterministic Audit, Confidence Scoring
AGENTIC AI · LLM EVAL · HEALTHCARE-GRADE

Python · LangGraph (8-agent) · FastAPI · ChromaDB · TF-IDF · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Triage → Retrieval → Config → Historical Incident → StructuredData → Reasoning → Validation → Human Escalation pipeline with durable state, conditional routing, and confidence-based HITL. Core rule enforced in code: LLM extracts and reasons; human or deterministic logic makes every final regulated call. Evaluated via AgentProbe: **93.6% precision, 87.9% F1, pass@3 88.3%** across 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants).

AgentProbe: Open-Source AI Evaluation Toolkit — 5-Metric Engine, CI/CD Gate, Model Routing
LLM MONITORING · PRODUCTION AI

Python · FastAPI (15 endpoints) · LangGraph · OpenAI · Anthropic · Groq · Ollama · ChromaDB · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- Fully local, zero-cost AI agent evaluation toolkit: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama, and offline + online evaluation loops. Used to evaluate Leasel across 1,240+ compliance test cases.

AI on FHIR: NLP Clinical Platform — EHR Integration, HIPAA RBAC, Multi-Language CLINICAL AI FHIR · HIPAA

Flask · LLMs · FHIR API · BERT · Embeddings · spaCy · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in manual clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR EHR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC with data isolation across structured medical records. React frontend, Docker containerized.

CrashClarity: Large-Scale Road Accident Analysis (104K Records) ANALYTICS ENGINEERING

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Tableau · GitHub Pages — Portfolio

- **85%+ accuracy** — end-to-end pipeline: ETL, feature engineering across weather/road/time variables, Random Forest + Logistic Regression, interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

AI systems, ML modeling, data engineering, NLP, statistical modeling, cloud infrastructure, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, LLM Orchestration, Tool Calling, Function Calling, Structured Outputs, Orchestration Runtime, Durable State Management, HITL, Human-in-the-Loop, RAG Pipelines, Hybrid Search, Semantic Search, Vector Databases, ChromaDB, Qdrant, Embeddings, Citation Grounding, LLM Evaluation, LLM-as-Judge, Golden Datasets, Regression Harnesses, Offline Evaluation, Online Evaluation, A/B Testing, Model Routing, Cost Controls, Hallucination Detection, Confidence Scoring, Guardrails, Auditability, Policy Compliance, Agent Observability, Trace-Level Observability, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Context Engineering, OCR, Document Intelligence, Production AI, LLMOps, OpenAI API, Anthropic API, Groq, Ollama, MCP

Clinical & Healthcare AI: Healthcare AI, Clinical AI, Ambient AI, EHR Integration, FHIR, HIPAA Compliance, PHI Compliance, Clinical NLP, Clinical Workflows, Patient Workflows, Revenue Cycle Management (RCM), Regulatory Compliance

Languages: Python, PHP, JavaScript, TypeScript, Java, Scala, SQL, PL/SQL, R, Bash, HTML/CSS, jQuery, Angular, React.js, Spring Boot, .NET Framework, Laravel

Backend & Systems: FastAPI, REST APIs, RabbitMQ, Redis, Microservices, Distributed Systems, Idempotency, Async Processing, Multi-Tenant Architecture, Data Isolation, API Design, Service Decomposition, Financial Settlement, Payments Integration, Entitlements Systems, Account State Management

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure (ADF, ADLS, Databricks, App Service), GCP, Docker, Kubernetes (K8s), OpenShift, CI/CD (GitHub Actions), Terraform, Infrastructure-as-Code (IaC), Prometheus, Grafana, OpenTelemetry, Apache Spark, PySpark, Git, Linux, Agile/Scrum, Jira

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, Isolation Forest, Random Forest, Logistic Regression, Anomaly Detection, Feature Engineering, Statistical Modeling, HuggingFace, BERT, spaCy, NLP, NumPy, Pandas, Matplotlib, Seaborn, Plotly, Tableau, Power BI, ggplot2

Data Engineering: ETL/ELT Pipelines, Data Warehouse Design, Dimensional Modeling, Medallion Architecture, Lakehouse Architecture, Star Schema, Data Quality Frameworks, Data Lineage, Data Governance, Financial Reconciliation, KPI Pipeline Design, Performance Tuning (CTE, Window Functions, Query Optimization), PostgreSQL, Snowflake, Teradata, MySQL, Oracle, SQL Server (T-SQL), Solr, NoSQL